Final Project Instructions

Project Overview

In this project we will have two challenges. They will both involve learning from limited image data. A big part of this assignment will be properly reporting your approaches and development stages. The testbed uses Vision Transformers (ViTs) as examples. You may also consider other approaches within the constraints below.

Teams

You will work in teams of 2-4 students. The work expectation will be proportional to your team size. Your final report should include a short discussion of what each team member contributed.

Challenge 1: Learning from Limited Data

Few-sample learning on CIFAR-10 images. We will consider a dataset with only 50 samples per class and 2 object classes. For testing, a larger dataset will be used (e.g. the CIFAR-10 test set).

Your job will be to make an algorithm that will train on 2 randomly selected classes from the CIFAR-10 training set with 50 samples each. You may not use any model trained on external data or any external data for your answer. You may consider any kind of approach, including non-deep learning methods and combinations of feature extraction, as long as it only uses the 50 examples per class for training.

The testbed provides a small ViT trained from scratch on the original 32-by-32 images. This is a starting point, not a requirement that a ViT outperform other methods.

Final evaluation: You will be provided several files with splits of training and test data. Run your final methods on them to report the final results.

Notes

1. The initial testbed has a separate development set for measuring progress and selecting hyperparameters. Do not train on these images. The final training script will receive just 50 samples per class. Do not use the final test data to select methods or hyperparameters.

2. Keep training and inference time reasonable and report both. If your best method requires extensive computation, also provide a faster alternative and discuss the tradeoff.

3. Accuracy can vary substantially in this setting. Evaluate on multiple sampled subsets before concluding that one approach is better than another.

Challenge 2: Learning with Limited Data Using External Data

Consider the same exercise, but now with the ability to use external data or models not trained on CIFAR-10, or to train your own model on external data. You may not train on additional CIFAR-10 images beyond the 50 samples per class, or use models trained on those additional images.

To bound the scope of this exercise, these are suggested approaches:

• Take a ViT pretrained on ImageNet, as illustrated in the testbed.

• Experiment with different ways to fine-tune it. In the testbed, the backbone is frozen and only a new classification head is trained. You could instead fine-tune the last transformer block or more of the model.

• Experiment with different fine-tuning optimization schedules.

The example checkpoint is **vit_tiny_patch16_224.augreg_in21k_ft_in1k** from timm. It was pretrained on ImageNet-21k and fine-tuned on ImageNet-1k. Use its supplied image preprocessing. You may explore other models, but document their pretraining data and respect the restriction on CIFAR-10.

If desired, you may consider more advanced techniques, for example meta-learning, semi-supervised learning with an external dataset, or other related ideas. You may also train your own pretrained model.

Notes

1. You should not use any CIFAR-10 images for training besides the 50 samples per class provided for that problem instance.

2. There is no single right answer. Many approaches are reasonable, as in Challenge 1.

3. If you are unsure whether an idea violates the intended constraint, ask the TA or instructor.

4. You are free to pretrain your own models on allowed external data sources.

Requirements / Evaluation

Each team will submit its best working solution evaluated on the final testing sets. Final submission and leaderboard instructions will be provided separately. You will be evaluated as follows:

Criterion	Percentage
Performance	20%
Report (excluding literature review)	70%
Originality	10%

Performance

Performance will take into account the overall accuracy of your model on the CIFAR-10 test set and any additional evaluation data provided by the instructor, as well as inference and training time. If your model is particularly slow, you may also submit a second model and discuss the tradeoffs. A portion of this credit will be awarded for exceeding a baseline performance threshold specified with the final evaluation instructions.

Report / Research Quality (excluding literature review)

1. Include details of the methods and strategies you tried, why they were appropriate, and the results of each method in your validation framework.

2. Evaluate at least 2 high-level methods for Challenge 1. Add 1 method for teams of more than 2 members. For example, a team of 3 should try at least 3 methods. These are minimums; you may study additional approaches.

3. Evaluate at least 1 high-level method for Challenge 2. Add 1 method for teams of more than 2 members. For example, a team of 3 should try at least 2 methods.

4. Explore appropriate hyperparameters and improvements in detail. For example, a ViT trained from scratch with cross-entropy is one method. Varying its depth, patch size, or training schedule would generally count as exploring that method, rather than as separate high-level methods. Another approach could use a CNN or a non-deep learning method such as logistic regression or gradient boosting. The distinction between methods can be ambiguous; ask the instructor or TA if unsure. The goal is to explore your approaches in detail and consider more than one general approach. You need not explore every direction equally deeply; you may focus on refining a particular strategy.

5. The size of the team will be taken into consideration when evaluating the depth and breadth of the approaches studied.

6. Explain how you validated your model and selected hyperparameters.

7. Present results clearly, with legible, clearly labelled figures. The appropriateness and clarity of your presentation will be considered.

8. Discuss your observations about the tradeoffs between training time, inference time, and performance.

Deliverables

Work in a group of 2-4 students. Include the names and student IDs of all group members with your submission.

1. **Report (PDF):** Describe your methods for both challenges, experimental setup, hyperparameter selection, results, and conclusions. Include the literature review and a short statement of each member's contributions, following the requirements above.

2. **Code / notebook:** Submit the code used for your experiments and final evaluation, together with the dependencies and instructions needed to run it.

3. **Video presentation:** Present the problem, your main approaches, key results, and conclusions. Identify all group members.

Literature Review / Originality

Provide a discussion of at least 1 page in total explaining where you sourced your methods and why you chose them. Sources may include course material, journal or conference papers, blog posts, unpublished articles, Kaggle forums, and other relevant material. Document where your ideas came from. To receive full credit for originality, at least one method you explore should go beyond what we explicitly covered in the course.

Computing Resources

Use Colab or an available GPU machine to prototype your results. The scratch example uses a small model and 32-by-32 images; the pretrained example uses 224-by-224 inputs. For a frozen backbone, you may cache image features to avoid recomputing them during classifier training. If longer runs are needed, ask about access to the lab GPU machines.

Alternative Project (must be pre-approved)

You may choose a research project of your own in deep learning. This option is largely intended for students already doing related research or applying deep learning to a suitable problem. Do not reuse research projects you have already finished, but you may work on complementary work. If interested, submit a preliminary one-paragraph proposal by email and obtain approval before starting.

Example Model References

Dosovitskiy et al., *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale.* https://arxiv.org/abs/2010.11929

ViT-Tiny pretrained checkpoint and preprocessing:
timm/vit_tiny_patch16_224.augreg_in21k_ft_in1k